# SIMATS ENGINEERING

# CSA6303- Threat Intelligence and Network Security

# A1-Malware Analysis Assignment

Assessment Questions

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|---|---|---|---|
| **Subject** | Threat Intelligence | **Total Marks** | 50 |
| **Questions** | 10 (all compulsory) | **Marks per Question** | 5 |
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| **Faculty** | Dr.V.Parthipan | **Slot** | A |

*Instructions: Answer all questions. Each question carries 5 marks. Support answers with examples and diagrams where appropriate.*

***CO3:*** Perform threat analysis using intelligence models and utilize industry tools for monitoring and detection.

***Code of Conduct:***

*I _S.RESHMA(192465027)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

## Signature of Student S.RESHMA

## Questions :

**1. Malware Analysis Using Static Techniques:**A security analyst receives a suspicious executable file attached to a phishing email. Before running the file, the analyst wants to examine its properties, file hash, strings, and imported functions. Explain how static malware analysis can help identify suspicious characteristics and assess the potential risk without executing the file.

**2. Malware Analysis Using Dynamic Techniques:**An organization suspects that an employee has downloaded a malicious program that may modify system files and communicate with an external server. Explain how dynamic malware analysis can be performed in an isolated virtual environment to observe the program's behavior and identify potential Indicators of Compromise (IoCs).

**3. Malware Investigation Using a Threat Intelligence Platform:** A company's security team discovers a suspicious file on an employee's computer. The file hash appears in a threat intelligence platform associated with a known malware family. Explain how the analyst can use threat intelligence sources to investigate the file, assess the reliability of the information, and recommend suitable containment measures.

**4. Malware Detection Through Data Collection and Correlation:** A university detects unusual processes on several computers, suspicious DNS requests, and unexpected outbound network connections. Explain how an analyst can collect and correlate endpoint logs, DNS records, firewall logs, and file hashes to determine whether the computers are infected with malware.

**5. Malware Investigation Using SIEM Tools:**A financial institution's Security Information and Event Management (SIEM) system generates alerts about suspicious process execution, repeated failed logins, and unusual network connections from the same workstation. Describe how the security analyst can use SIEM capabilities to correlate these events, investigate the possible malware infection, and prioritize the incident.

**6. Malware Detection and Log Analysis Using Splunk:** A company uses Splunk to monitor its network and endpoint logs. The security team suspects that malware is attempting to contact an external command-and-control server. Explain how Splunk can be used to search relevant logs, identify suspicious communication patterns, correlate events, and document evidence of possible malware activity.

**7. Malware Investigation Using the ELK Stack:**An e-commerce company uses the ELK Stack (Elasticsearch, Logstash, and Kibana) to monitor web server logs, firewall records, and endpoint events. The security team notices repeated connections to suspicious domains and unusual file activity. Explain how the ELK Stack can support malware investigation through log collection, processing, searching, visualization, and correlation.

**8. Threat Hunting for Malware Activities:** An organization's security team suspects that malware has infected an employee workstation but has not received any antivirus alerts. The workstation shows unusual CPU usage, unfamiliar processes, and connections to unknown external IP addresses. Describe a threat-hunting approach to investigate the workstation, identify suspicious behaviors, collect relevant IoCs, and recommend remediation measures.

**9. Comparing Malware Analysis Tools Used in the IT Industry:** A cybersecurity team needs to investigate a suspicious file and determine whether it contains malicious code. Compare static analysis tools, dynamic analysis tools, network traffic analyzers, and threat intelligence platforms in terms of their purpose and usefulness. Recommend an appropriate combination of tools for conducting a malware investigation.

**10. Malware Analysis Report and Incident Response:** A company discovers a suspicious executable on an employee's computer. Initial analysis identifies an unfamiliar process, a modified system configuration, and communication with a suspicious external domain. Prepare an outline for a malware analysis report that includes file information, analysis methodology, observed behavior, IoCs, threat intelligence findings, risk assessment, and recommended containment and recovery actions.

## Answers

## 1. Malware Analysis Using Static Techniques [10 Marks]

**Static malware analysis examines a file without executing it, allowing analysts to identify suspicious characteristics safely.**

### Steps in Static Analysis

1. **Examine File Properties**
    - **Check file name, size, type, creation date, and digital signature.**
    - **Identify whether the file claims to be legitimate software.**
2. **Calculate File Hashes**
    - **Generate MD5, SHA-1, or SHA-256 hashes.**
    - **Compare hashes with malware databases such as VirusTotal.**
3. **Extract Strings**
    - **Analyze readable text within the executable.**
    - **Look for:**
        - **Suspicious URLs**
        - **IP addresses**

        - **File paths**
        - **Registry keys**
        - **Commands**

4. **Analyze Imported Functions**
    - **Identify APIs used by the malware.**
    - **Functions such as:**
        - **CreateProcess()**
        - **WriteFile()**
        - **RegSetValue()**
        - **InternetOpen()**
    - **indicate possible malicious behavior.**
5. **Inspect PE Header**
    - **Check compilation time and entry points.**
    - **Detect packing or obfuscation techniques.**

**Benefits**

- **Safe analysis without infection risk.**
- **Early identification of malware characteristics.**
- **Collection of Indicators of Compromise (IoCs).**

**Conclusion**

**Static analysis helps security analysts assess a suspicious file's risk by examining its structure, hashes, strings, and imported functions without running the file.**

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## 2. Malware Analysis Using Dynamic Techniques [10 Marks]

**Dynamic analysis involves observing malware behavior during execution in an isolated environment.**

**Steps**

**1. Create an Isolated Sandbox**

- **Virtual Machine (VM)**
- **Sandbox Environment**
- **No access to production systems**

**2. Execute the Program**

**Monitor the malware while it runs.**

**3. Observe System Changes**

**Check for:**

- **New files created**
- **Registry modifications**
- **System configuration changes**
- **Added services**

**4. Monitor Network Activity**

**Identify:**

- **Outbound connections**
- **DNS queries**
- **Command-and-Control (C2) communication**
- **Data exfiltration attempts**

**5. Analyze Processes**

**Observe:**

- **Process creation**
- **Process injection**
- **Privilege escalation**

**Indicators of Compromise (IoCs)**

- **Malicious domains**
- **Suspicious IP addresses**
- **Registry keys**
- **File hashes**
- **New scheduled tasks**

**Conclusion**

**Dynamic analysis reveals real-time malware behavior and helps identify IoCs that may not be visible during static analysis.**

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**3. Malware Investigation Using a Threat Intelligence Platform [10 Marks]**

**A threat intelligence platform provides information about known threats and malware families.**

**Investigation Process**

**Step 1: Search the File Hash**

- **Compare the file hash against intelligence databases.**
- **Determine whether it is associated with known malware.**

**Step 2: Verify Threat Information**

**Collect:**

- **Malware family name**

- **Threat actor information**
- **Attack techniques**
- **Associated IoCs**

**Step 3: Assess Reliability**

**Validate findings using:**

- **Multiple intelligence sources**
- **Vendor reputation**
- **Confidence ratings**
- **Recent threat reports**

**Step 4: Determine Impact**

**Analyze:**

- **Affected systems**
- **Potential data loss**
- **Scope of infection**

**Containment Measures**

1. **Isolate infected endpoints.**
2. **Block malicious IPs and domains.**
3. **Remove malicious files.**
4. **Reset compromised credentials.**
5. **Increase monitoring.**

**Conclusion**

**Threat intelligence helps analysts verify malware, understand risks, and implement effective containment strategies.**

## 4. Malware Detection Through Data Collection and Correlation [10 Marks]

**Analysts can identify malware by collecting and correlating information from multiple security sources.**

**Data Collection Sources**

**Endpoint Logs**

- **Running processes**
- **Application activity**
- **Security events**

**DNS Logs**

- **Domain requests**
- **Suspicious lookups**

**Firewall Logs**

- **Inbound and outbound traffic**
- **Blocked connections**

**File Hashes**

- **Verify suspected files against threat databases**

**Correlation Process**

| Source | Information |
|---|---|
| Endpoint Logs | Suspicious process execution |
| DNS Logs | Requests to malicious domains |
| Firewall Logs | Connections to suspicious IPs |

| | |
|---|---|
| **File Hashes** | **Known malware signatures** |

**Analysis**

- **Identify systems showing all indicators.**
- **Determine infection timeline.**
- **Identify malware communication channels.**

**Conclusion**

**Correlating logs from multiple sources provides stronger evidence of malware infection than relying on a single data source.**

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**5. Malware Investigation Using SIEM Tools [10 Marks]**

**A SIEM helps analysts detect malware by aggregating and correlating security events.**

**Investigation Steps**

**1. Collect Logs**

**Gather data from:**

- **Endpoints**
- **Firewalls**
- **Servers**
- **Authentication systems**

**2. Correlate Events**

**Example:**

| Event | Observation |
| --- | --- |
| Failed Logins | Multiple attempts |
| Suspicious Process | Unknown executable |
| Network Activity | Unusual external connection |

**The combination may indicate malware infection.**

**3. Timeline Analysis**

**Determine:**

- **Initial infection**
- **User activity**
- **Malware execution**
- **Network communication**

**4. Threat Prioritization**

**Assess:**

- **Severity**
- **Criticality of affected asset**
- **Business impact**

**Benefits**

- **Centralized visibility**
- **Faster investigation**
- **Improved incident response**

**Conclusion**

**SIEM tools help correlate suspicious events, identify malware activity, and prioritize incidents based on risk.**

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**6. Malware Detection and Log Analysis Using Splunk [10 Marks]**

**Splunk enables real-time monitoring and analysis of malware-related activities.**

**Investigation Process**

**Search Endpoint Logs**

**Identify:**

- **Unknown processes**
- **Application crashes**
- **Unauthorized software execution**

**Analyze Network Logs**

**Search for:**

- **Connections to external IPs**
- **Repeated DNS lookups**
- **Beaconing patterns**

**Correlate Events**

**Example:**

1. **Suspicious process starts.**
2. **DNS request to unknown domain.**
3. **Outbound traffic to external server.**

**Together these indicate possible malware activity.**

**Create Dashboards**

**Visualize:**

- **Infected hosts**
- **Alert frequency**
- **Communication patterns**

**Preserve Evidence**

**Document:**

- **Timestamps**
- **Log entries**
- **IP addresses**
- **File hashes**

**Conclusion**

**Splunk helps analysts search logs, correlate events, visualize suspicious activities, and document evidence for malware investigations.**

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**7. Malware Investigation Using the ELK Stack [10 Marks]**

**The ELK Stack consists of:**

- **Elasticsearch (Storage and Search)**
- **Logstash (Log Processing)**
- **Kibana (Visualization)**

**Investigation Process**

**Log Collection**

**Logstash collects:**

- **Web server logs**

- **Firewall logs**
- **Endpoint logs**

**Processing**

**Logstash:**

- **Parses data**
- **Removes inconsistencies**
- **Normalizes fields**

**Searching**

**Elasticsearch enables:**

- **Fast searches**
- **IOC queries**
- **Timeline analysis**

**Visualization**

**Kibana provides:**

- **Dashboards**
- **Graphs**
- **Malware activity trends**

**Correlation**

**Analysts correlate:**

- **Suspicious domains**
- **File modifications**
- **Endpoint activity**
- **Network connections**

**Conclusion**

**The ELK Stack helps security teams collect, process, search, visualize, and correlate logs for effective malware investigations.**

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**8. Threat Hunting for Malware Activities [10 Marks]**

**When antivirus does not generate alerts, threat hunting is used to proactively search for hidden malware.**

**Investigation Steps**

**1. Analyze System Performance**

**Check:**

- **CPU usage**
- **Memory usage**
- **Disk activity**

**2. Investigate Running Processes**

**Identify:**

- **Unknown executables**
- **Unsigned applications**
- **Suspicious services**

**3. Examine Network Connections**

**Look for:**

- **External IP communications**
- **Unknown domains**
- **Frequent beaconing**

**4. Search for Persistence Mechanisms**

**Check:**

- **Startup folders**
- **Scheduled tasks**
- **Registry run keys**

**5. Collect IoCs**

**Gather:**

- **File hashes**
- **Domains**
- **IP addresses**
- **Process names**

**6. Verify Using Threat Intelligence**

**Compare observed IoCs with threat intelligence databases.**

**Remediation**

- **Isolate workstation.**
- **Remove malware.**
- **Change passwords.**
- **Patch vulnerabilities.**
- **Deploy EDR monitoring.**

**Conclusion**

**Threat hunting identifies hidden malware through behavioral analysis, IoC collection, and proactive investigation techniques.**

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**9. Comparing Malware Analysis Tools Used in the IT Industry [10 Marks]**

| Tool Category | Purpose | Advantages |
|---|---|---|
| Static Analysis Tools | Examine files without execution | Safe and quick |
| Dynamic Analysis Tools | Monitor runtime behavior | Reveals actual actions |
| Network Traffic Analyzers | Inspect network communications | Detects C2 traffic |
| Threat Intelligence Platforms | Verify known threats | Provides threat context |

**Examples**

**Static Analysis Tools**

- **PEStudio**
- **Strings**
- **ExifTool**

**Dynamic Analysis Tools**

- **Cuckoo Sandbox**
- **Any.Run**
- **VMware Sandbox**

**Network Traffic Analyzers**

- **Wireshark**
- **Zeek**
- **TCPDump**

**Threat Intelligence Platforms**

- **VirusTotal**
- **AlienVault OTX**
- **MISP**

**Recommended Combination**

1. **Static Analysis → Initial inspection.**
2. **Dynamic Analysis → Behavior analysis.**
3. **Network Analysis → Communication monitoring.**
4. **Threat Intelligence → Threat verification.**

**Conclusion**

**Combining all four tool categories provides comprehensive malware investigation and accurate threat assessment.**

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**10. Malware Analysis Report and Incident Response [10 Marks]**

**Malware Analysis Report**

**1. Executive Summary**

**A suspicious executable was discovered on an employee workstation and exhibited potentially malicious behavior.**

**2. File Information**

- **File Name**
- **File Size**
- **File Type**
- **MD5/SHA256 Hash**
- **Detection Date**

**3. Analysis Methodology**

**Static Analysis**

- **Hash analysis**
- **String analysis**
- **Import analysis**

**Dynamic Analysis**

- **Sandbox execution**
- **Behavior monitoring**

**4. Observed Behavior**

- **Unknown process execution**
- **Registry modifications**
- **System configuration changes**
- **External network communication**

**5. Indicators of Compromise (IoCs)**

| Category | Indicator |
|---|---|
| File | SHA256 Hash |
| Process | Suspicious Executable |
| Network | Malicious IP Address |
| Domain | Suspicious Domain |
| Registry | Persistence Key |

**6. Threat Intelligence Findings**

- **Malware family identified.**

- **Associated attacker techniques.**
- **Known indicators matched.**

**7. Risk Assessment**

| Factor | Assessment |
|---|---|
| Severity | High |
| Likelihood | High |
| Business Impact | Significant |

**8. Containment Actions**

- **Isolated infected host.**
- **Blocked malicious domains.**
- **Disabled compromised accounts.**

**9. Recovery Actions**

- **Malware removal.**
- **Password reset.**
- **System restoration.**
- **Security patch deployment.**

**10. Recommendations**

1. **Enable EDR solutions.**
2. **Enforce Multi-Factor Authentication (MFA).**
3. **Improve email filtering.**
4. **Conduct regular threat hunting.**
5. **Provide security awareness training.**

## Conclusion

**The malware analysis report documents the investigation process, identified IoCs, observed behavior, risk assessment, and remediation actions required to contain the threat and restore secure operations.**